

Lesson 6

Dynamics of continuous systems – axial vibrations of beams

1 Assumptions and equation of motion

We will treat the problem of axial beam vibrations under the two following main assumptions:

- 1) Linear elastic constitutive law: we will consider a linear, isotropic relationship between stress and strain in the beam material; this assumption is normally well satisfied by metallic materials (steel, aluminium) when stresses and deformations in the material are kept well below the plastic yield limit.
- 2) constant section and homogeneous material; as explained in section 2.1, we will consider the section of the beam as constant in shape and dimensions along the beam axis. Furthermore, we will assume that all physical properties of the beam material (density, modulus of elasticity) do not depend upon the particular position inside the beam volume.

We consider then a beam like the one represented in Figure 1.1. We choose a Cartesian reference with the x axis coinciding with the axis of the beam, while the y and z axis lie in the plane containing the section of the beam. Though this is not relevant to the subject treated in this lesson (but instead to the one treated in the next lesson, bending vibration of beam systems), we choose the y and z axis to coincide with the principal axis of the beam section (the reader is referred to any book of solid mechanics for a definition of the section's principal axis). Correspondingly, we use symbols u , v , w to denote the displacement components of the generic material point in the beam along the x , y and z directions respectively.

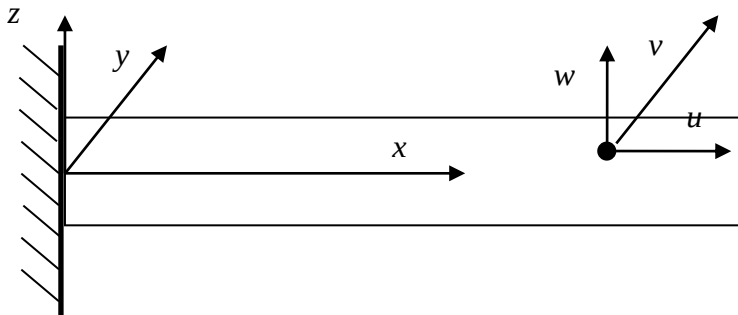


Figure 1.1 Cartesian reference and scalar components of displacements.

When dealing with the axial vibrations of the system depicted in Figure 1.1, we assume that all external forces on the system are acting along the beam axis, so that the internal forces in the beam will have only one non-zero component, the axial force (or traction) N . After the “De Saint Venant’s” solution of the elastic problem for the case of beam traction in a uniform beam, we may state that in presence of an axial internal force N (positive if traction, see Figure 1.2) a uniform distribution of tensional stresses σ_x will develop on each section of the beam¹ with:

¹ For the sake of preciseness, we should recall this solution is valid only at a sufficient distance from the points where external forces and/or constraints are applied to the beam. However, we will apply it to the whole beam

$$\sigma_x = \frac{N}{A} \quad (1.1)$$

where A is the area of the beam section, while all other stress components in the beam material (tensional stresses σ_y and σ_z and shear stresses τ_{xy} , τ_{yz} , and τ_{zx}) will be zero everywhere.

Due to the linear elastic constitutive law assumed for the material in the beam, the following components of strain will arise:

$$\varepsilon_x = \frac{\sigma_x}{E} = \frac{N}{EA} \quad ; \quad \varepsilon_y = \varepsilon_z = \frac{-\nu\sigma_x}{E} = \frac{-\nu N}{EA} \quad (1.2)$$

where ε_x , ε_y , and ε_z are the three dilatation components of strain, E and ν are respectively the modulus of elasticity (Young modulus) and the Poisson coefficient of the beam material, while the three components of shear deformation (γ_{xy} , γ_{yz} , and γ_{zx}) are all zero, due to the fact that no shear stress is present in the material.

If we neglect the contraction of the beam section that is produced by the presence of transversal dilatations ε_y , and ε_z (normally negligible with respect to the effect of longitudinal dilatations ε_x) we may then assume that the material in the beam will be only subjected to longitudinal displacements u , that will be the same for all point belonging to the same section in the beam (Which means, for all points having the same y and z coordinates). Thus, the displacements in the beam will be represented by a function u representing axial motions in the beam as function of the x spatial coordinate and additionally of time:

$$u = u(x, t) \quad (1.3)$$

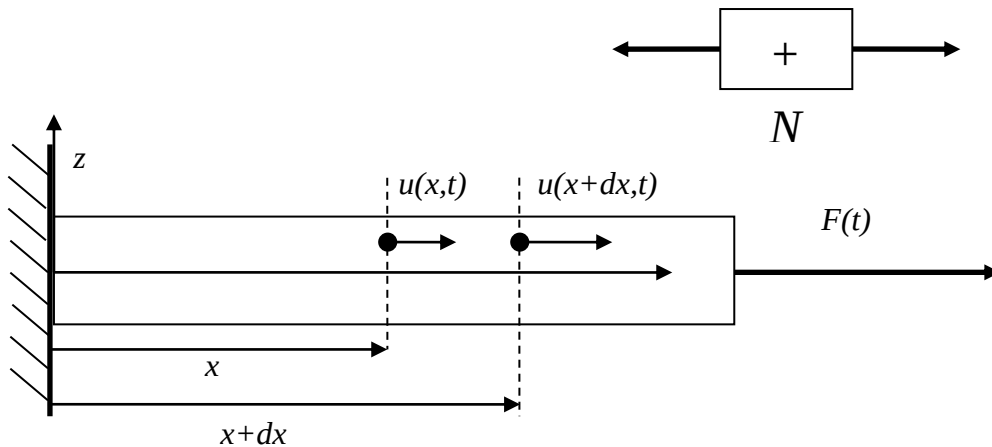


Figure 1.2 Components of internal force and displacement in the study of axial vibrations of beams

The normal strain ε_x is defined as the ratio between the elongation of fibres lying along the x direction:

$$\varepsilon_x = \frac{\Delta \ell_x}{\ell_x} = \frac{u(x+dx) - u(x)}{dx} = \frac{u(x) + \frac{\partial u}{\partial x} dx - u(x)}{dx} = \frac{\partial u}{\partial x} \quad (1.4)$$

so that, by introducing (1.4) into the first of equations (1.2) and rearranging the result to find an expression for the internal force component N we get:

$$N = EA\varepsilon_x = EA\frac{\partial u}{\partial x} \quad (1.5)$$

Now, in order to write the equation of motion for the axial vibrations of a beam, we take the piece of beam enclosed between the two sections placed at the longitudinal positions x and $x+dx$ (see Figure 1.2) and we write a dynamic equilibrium equation, according to the D'Alembert's principle². The forces (including inertia forces) acting on the considered infinitesimal piece of the beam are sketched in figure 2.3 and consist of:

- axial force $N(x)$ acting on the left face of the element;
- axial force $N(x+dx)$ acting on the right face of the element;
- inertia force, expressed as the product of the infinitesimal mass of the element dm times the acceleration of the element $\frac{\partial^2 u}{\partial t^2}$, with the sign changed.

the axial forces are disposed on the sketch of figure 3.3 according to the convention on axial forces introduced in Figure 1.2 (force positive if traction), while the element acceleration is positive when pointing rightwards, since this is the convention chosen for u displacements.

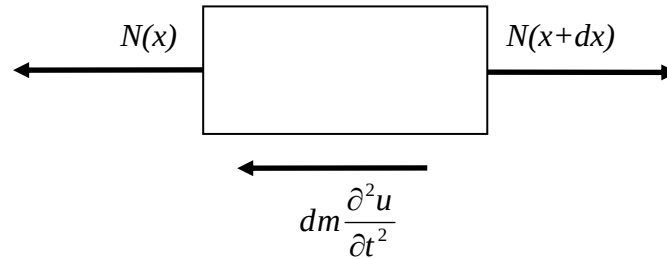


Figure 1.3 Equilibrium of a beam element with infinitesimal dimension dx .

The application of D'Alembert principle brings:

$$-N(x) + N(x+dx) = dm\frac{\partial^2 u}{\partial t^2}$$

where the infinitesimal mass dm of the element may be expressed as the product of material density ρ times the infinitesimal element volume $A dx$:

$$dm = \rho A dx = m dx \quad \text{with} \quad m = \rho A$$

where parameter m takes the name of “linear mass” or “mass per unit length” of the beam, and owing to our assumptions (constant beam section, uniform material density) may also be obtained as the ratio between the total mass of the beam and its total length. We then get:

$$-N(x) + N(x+dx) = -N(x) + N(x) + \frac{\partial N}{\partial x} dx = m dx \frac{\partial^2 u}{\partial t^2}$$

and, by proper simplification and re-arranging:

$$\frac{\partial N}{\partial x} = m \frac{\partial^2 u}{\partial t^2}$$

finally, by recalling (1.5) we get:

² This means an equation stating that the sum of all forces acting on the infinitesimal piece of beam along the axis direction, including inertia forces must be zero.

$$EA \frac{\partial^2 u}{\partial x^2} = m \frac{\partial^2 u}{\partial t^2} \quad (1.6)$$

which represents the equation of motion for the axial vibrations of the beam. We note that while for a discrete (n -dof) system the set of independent coordinates was represented by n functions of time ($x_1(t)$, $x_2(t)$, ..., $x_n(t)$), for a continuous system as the one treated in this lesson we had to take as independent kinematical parameter one single function $u(x,t)$, depending upon both the spatial coordinate x and the time coordinate t . If we consider that the axial displacement function u is defined for infinite values of coordinate x , spanning the whole length of the beam, we must recognise that introducing function $u(x,t)$ as the independent coordinate for the beam corresponds to introducing an infinite number of time dependent coordinates (namely, one for each of the possible infinite values of coordinate x). Correspondingly, instead than a set of n differential equations where the derivatives wrt time of $x_1(t)$, $x_2(t)$, ..., $x_n(t)$ appear, we get in this case one single partial derivatives differential equation. In some sense, the passage from a discrete to a continuous system may then be regarded as the limit case where the number n of degrees of freedom of the discrete system tends to infinity.

Before closing this section, we observe that eq. (1.6) may be rewritten in the form:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (1.7)$$

with:

$$c^2 = \frac{EA}{m} \quad ; \quad c = \sqrt{\frac{EA}{m}} \quad (1.8)$$

the form (1.7) of the equation of represents an entire class of differential problems, such as axial vibrations of beams (the case considered here), torsional vibration of beams, transversal vibrations of a stretched string, sound propagation in a mono-dimensional propagation mean.

Parameter c takes here the meaning of the *speed of propagation* of an axial perturbation inside the beam. Owing to its definition (1.8) and to the expression of the mass per unit length, it may be observed that this speed is independent from the actual size and shape of the beam section, and only depends upon material properties, since:

$$c = \sqrt{\frac{EA}{m}} = \sqrt{\frac{EA}{\rho A}} = \sqrt{\frac{E}{\rho}}$$

2 Axial vibrations of beams: propagative solutions

In this section we start discussing the solution of eq. (1.7). It may be shown that the general solution takes the form:

$$u(x,t) = f_1(x-ct) + f_2(x+ct) \quad (2.1)$$

where f_1 and f_2 are two arbitrary functions, each one being twice derivable. We easily recognise that expression (32.1) is solution of equation (1.7) since (by denoting as f'' the second derivative of functions f_1 and f_2):

$$\frac{\partial^2 u(x,t)}{\partial x^2} = f_1''(x-ct) + f_2''(x+ct) \quad ; \quad \frac{\partial^2 u(x,t)}{\partial t^2} = c^2 f_1''(x-ct) + c^2 f_2''(x+ct)$$

In order to provide a physical meaning for solution (3.1) to problem (2.7) we define:

$$s_1 = x - ct \quad ; \quad s_2 = x + ct$$

then, we concentrate initially on the first term of solution (2.1), as if it was $f_2=0$. We may observe in this case that $f_1(s_1)$ represents a waveform (with arbitrary shape) defined in terms of the auxiliary coordinate s_1 . If we consider time $t=0$, then the auxiliary coordinate s_1 coincides with the space coordinate x , so that we may identify function f_1 with the axial deformation of the beam at the initial time $t=0$ (under the above assumption $f_2=0$). If we then consider a subsequent time $t>0$, the relationship between s_1 and x becomes $x=s_1+ct$, so that the same value of coordinate s_1 representing the initial beam deformation in a given position s_1 is now obtained for $t>0$ in a different location that is displaced by a quantity ct in the positive direction of the x axis, which means rightwards according to the conventions previously selected. Thus, $f_1(x-ct)$ represents a waveform of axial deformation that is propagating rightwards inside the beam, while keeping the same waveform. This situation is depicted in the upper part of Figure 2.1.

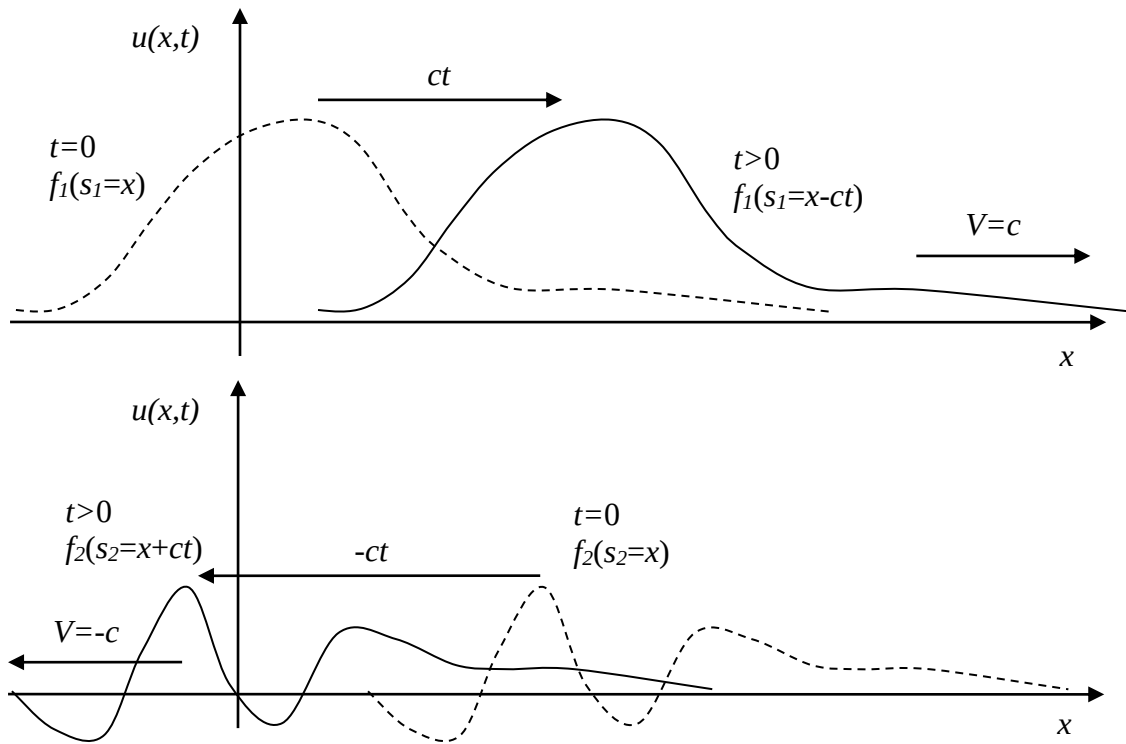


Figure 2.1 Propagation of two axial perturbations in the beam

Also the second part of solution (2.1), $f_2(x+ct)$, may be interpreted as the propagation of a second axial perturbation with steady waveform inside the beam, with the only difference that in this case, due to the different sign in front of the ct term, this second addendum represents a propagative term moving leftwards instead than rightwards.

Thus, we may conclude from this analysis that when an initial condition is assigned to the axial positions of the different beam sections at the initial time $t=0$, two waves are “launched” in the beam, with opposite directions, both moving with an absolute value of the speed defined by eq. (1.8). Thanks to this physical interpretation, the solution (2.1) of equation (1.7) is called the “propagative solution” for the problem of axial vibrations in a beam.

3 Axial vibrations of beams: stationary solutions

The propagative solution (2.1) is useful to deal with problems concerning wave propagation in infinite media: this may be the case, e.g., for sound propagation into a supposedly infinite mono-dimensional medium. However, in the case of axial vibration in a beam with finite length, the two waves generated at $t=0$ will reach the ends of the beam and will be (at least partially) reflected, so that positive and negative interference will take place between the travelling and the reflected

waves. By decomposing the travelling waves into a Fourier series and considering properly the boundary conditions at the extremities of the beam, it may be demonstrated that this positive and negative interference will produce stationary waveforms of vibration, which means an axial vibration that may be described as a waveform fixed in space whose amplitude is modulated by a time-dependent coefficient. This kind of so-called “stationary vibration” may be expressed in the form:

$$u(x, t) = \alpha(x)\beta(t) \quad (3.1)$$

where $\alpha(x)$ is a function of space alone describing the waveform of the stationary vibration, while $\beta(t)$ represents a time dependent amplitude coefficient being applied on the spatial waveform $\alpha(x)$. Figure 3.1 shows the physical meaning of the stationary solution, assuming e.g. that $\beta(t)$ oscillates between ± 1 .

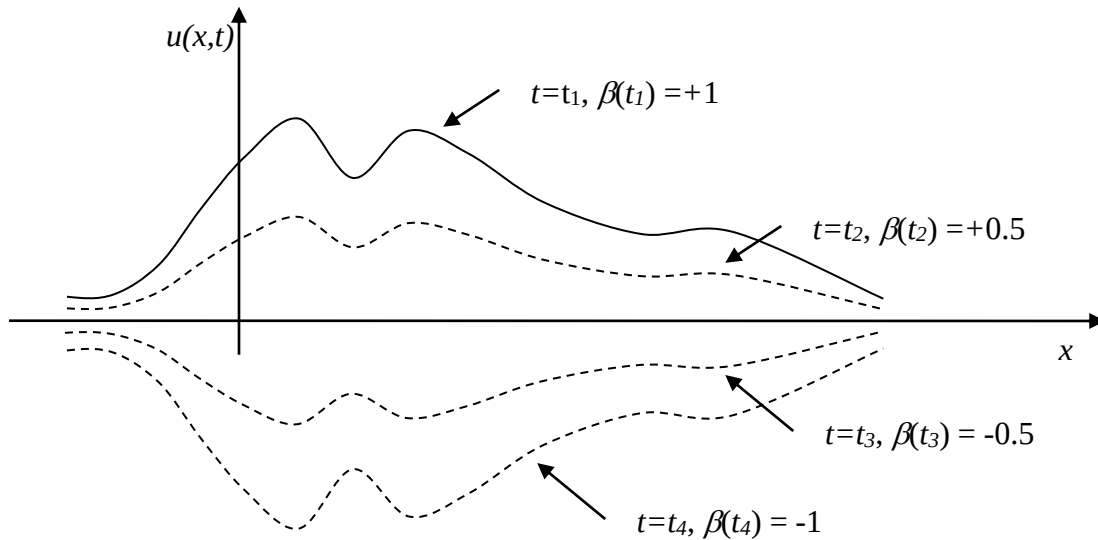


Figure 3.1 Interpretation of the stationary solution (4.1) for the axial vibration of the beam

The actual expressions for the two functions $\alpha(x)$ and $\beta(t)$ could be obtained from the propagative solution (3.1) by using a Fourier series expression in conjunction with a proper representation of the boundary conditions at the ends of the beam. However, we prefer to use here a much simpler approach, sometimes called “separation of variables”, which consists in replacing expression (3.1) into equation (1.7).

To this end, we observe that:

$$\frac{\partial^2 \alpha(x)\beta(t)}{\partial x^2} = \alpha''(x)\beta(t) \quad ; \quad \frac{\partial^2 \alpha(x)\beta(t)}{\partial t^2} = \alpha(x)\ddot{\beta}(t)$$

where $\alpha''(x)$ stands for the second derivative of α wrt x . By introducing the above relationships in eq. (1.7) we get:

$$\alpha(x)\ddot{\beta}(t) = c^2 \alpha''(x)\beta(t)$$

then, we separate at first side of the above equation all terms depending upon the time coordinate t , from the terms depending upon the space coordinate x that are placed at the right hand side. Since the expression at left hand side is time dependent while the one at the right hand side is space dependent, the only way to satisfy the equality sign is to assume that both terms are equal to the same constant. For reasons that will soon be clear, this constant must be negative valued, so that the constant expression $-\omega^2$ is used to make sure of the negative sign:

$$\frac{\ddot{\beta}(t)}{\beta(t)} = c^2 \frac{\alpha''(x)}{\alpha(x)} = -\omega^2 \quad (3.2)$$

We now use the above equation to derive one differential equation for the time dependent function $\beta(t)$ and one separate differential equation for the space dependent function $\alpha(x)$:

$$\ddot{\beta}(t) + \omega^2 \beta(t) = 0$$

$$\alpha''(x) + \left(\frac{\omega}{c}\right)^2 \alpha(x) = 0$$

the first equation (in time) has the well known form of the equation describing the free vibration of an undamped 1-dof system, and thus the solution takes the form of a harmonic function of time, that may be represented as a cosine function with amplitude C and phase φ :

$$\beta(t) = C \cos(\omega t + \varphi) \quad (3.3)$$

From the above result the reason why we required in equation (3.2) the constant $-\omega^2$ to be negative valued should now be clear, since in case of positive sign, we would come out with a differential equation for the time dependent solution $\beta(t)$ having the form:

$$\ddot{\beta}(t) - \omega^2 \beta(t) = 0$$

that would have brought as a solution an exponential expression for $\beta(t)$ where one part of the solution would be exponentially increasing with time. That would have meant a “divergent” and thus unstable motion for our system, which is instead stable by nature.

The second equation (in space) defining the part of solution $\alpha(x)$ has actually the same form as the previous one, except for the fact that the independent variable is now x (space) instead than t (time) and for the fact that constant ω^2 is replaced by $(\omega/c)^2$. It is then easy to see that the corresponding solution will be in the form of a harmonic function of space with the ratio ω/c in the place of the constant ω . However, for the space dependent part of the solution $\alpha(x)$ we prefer (for reasons that will be soon clear) to use for the harmonic solution the form based on the linear combination of a cosine and a sine function (without phases) instead than the one based on a single cosine function with generic phase:

$$\alpha(x) = A \cos\left(\frac{\omega}{c} x\right) + B \sin\left(\frac{\omega}{c} x\right) \quad (3.4)$$

it is left to the reader to verify that the ω/c ratio has dimensions length at power -1, so that the argument of the sine and cosine functions in the above solution are pure numbers. From (3.1), (3.3) and (3.4) we get:

$$u(x, t) = \alpha(x)\beta(t) = \left[A \cos\left(\frac{\omega}{c} x\right) + B \sin\left(\frac{\omega}{c} x\right) \right] \cos(\omega t + \varphi) \quad (3.5)$$

the amplitude parameter C in Eq. (3.3) being now embedded in parameters A and B from Eq. (3.4). Expression (3.5) represents the general solution for the stationary free axial vibration of a beam with constant section, regardless the boundary conditions acting on the beam at its ends. In order to introduce the concept of natural frequencies and associated modes of vibration for the problem under study, we have to specify the boundary conditions applied to the beam.

In this section, we consider as an example the “clamped-clamped” combination of boundary conditions, which means that two constraints are placed at both ends of the beam, preventing the axial displacement of the ends of the beam.

If we choose $x = 0$ as the left end of the beam and $x = L$ as the right end (L being the total length of the beam), we may express the “clamped-clamped” boundary conditions as:

$$\begin{aligned} u(x,t)\Big|_{x=0} &= 0 \quad \text{for any time } t \\ u(x,t)\Big|_{x=L} &= 0 \quad \text{for any time } t \end{aligned}$$

replacing the first of these two boundary conditions in solution (3.5) we get:

$$A = 0 \quad (3.6)$$

where the time dependent expression has been simplified since the equality must hold for any time t . We now replace in (3.5) the second boundary condition, taking into consideration that $A=0$ we get:

$$B \sin\left(\frac{\omega}{c} L\right) = 0$$

that may be satisfied in two alternative ways, either by setting $B=0$, or by setting $\sin\left(\frac{\omega}{c} L\right) = 0$.

The first of the two solutions would imply $A=0$, $B=0$ and then no motion at all for the system (what is called the “trivial” solution). In order to get non trivial solutions, we must then require:

$$\sin\left(\frac{\omega}{c} L\right) = 0 \rightarrow \frac{\omega}{c} L = k\pi \quad k = 1, 2, \dots \quad (3.7)$$

from the above relationship, we conclude that the free axial vibration of the clamped-clamped beam is possible if the angular frequency parameter ω assumes any of the values defined by formula:

$$\omega_k = \frac{k\pi}{L} c = \frac{k\pi}{L} \sqrt{\frac{EA}{m}} = \frac{k\pi}{L} \sqrt{\frac{E}{\rho}} \quad k = 1, 2, \dots \quad (3.8)$$

where ω_k takes the meaning of the k -th *natural frequency* of the clamped-clamped beam in its axial vibrations. We observe that in (3.8) coefficient k has to assume integer values, in order to satisfy the boundary condition posed by the presence of a clamp at the right beam end. However, there is no upper limit posed on the value of k , which means that formula (3.8) provides an infinite sequence of natural frequencies for the continuous system, which is in line with our previous observation that a continuous system may be seen as the limit case of a n -dof system where $n \rightarrow \infty$. We will come back soon on this point.

In order to define the solution of motion associated with the k -th natural frequency of the beam, we now replace (3.6) and (3.7) into the general expression of stationary axial vibrations (3.5), and we get:

$$u_k(x,t) = B_k \sin\left(\frac{k\pi}{L} x\right) \cos(\omega_k t + \varphi_k) \quad (3.9)$$

which represents the k -th component of the stationary axial vibration of the beam, associated with the natural angular frequency ω_k . The general motion of the system will be a combination of all the possible components with expression (3.9), and will thus result into:

$$u(x,t) = \sum_{k=1}^{\infty} u_k(x,t) = \sum_{k=1}^{\infty} B_k \sin\left(\frac{k\pi}{L} x\right) \cos(\omega_k t + \varphi_k) \quad (3.10)$$

where the amplitude parameters B_k and the phase parameters φ_k have to be determined by considering a particular set of initial conditions for the system.

The space dependent part of the k -th vibration component (3.9), that is:

$$\alpha_k(x) = B_k \sin\left(\frac{k\pi}{L} x\right) \quad (3.11)$$

represents the shape of the beam vibration, and thus defines the k -th *mode of vibration*, associated with the natural angular frequency ω_k . As for the discrete systems, the mode of vibration is defined with an arbitrary amplitude B_k^3 , which may be conventionally set to 1, so that the modes of vibration become sine functions with wavelengths that are entire sub-multiple of a base wavelength equal to the double of the beam span. Figure 4.2 shows the shape of the first modes of vibration of the axial vibrations for a clamped–clamped beam: in the first mode of vibration all the sections are vibrating in phase, with the largest amplitude of vibration occurring at centre-span, in the second mode of vibration, the left and right halves of the beam are vibrating in counter-phase, with the maximum amplitude of displacement occurring at $\frac{1}{4}$ and $\frac{3}{4}$ of the span length, and with a *nodal section* at centre-span. We recall that a nodal point is a point that does not move in a specified mode of vibration of the system. The third and fourth modes of vibration introduce a larger number of nodes and parts of the beam moving in counter-phase each other, with all even modes (n.2, n.4 etc.) having a node in the central section. We note by the way that for increasing frequency, the associated modes of vibration become increasingly “winding”: this is a quite general rule that applies not only to axial vibration of beams, but in general to the vibration of discrete and continuous systems.

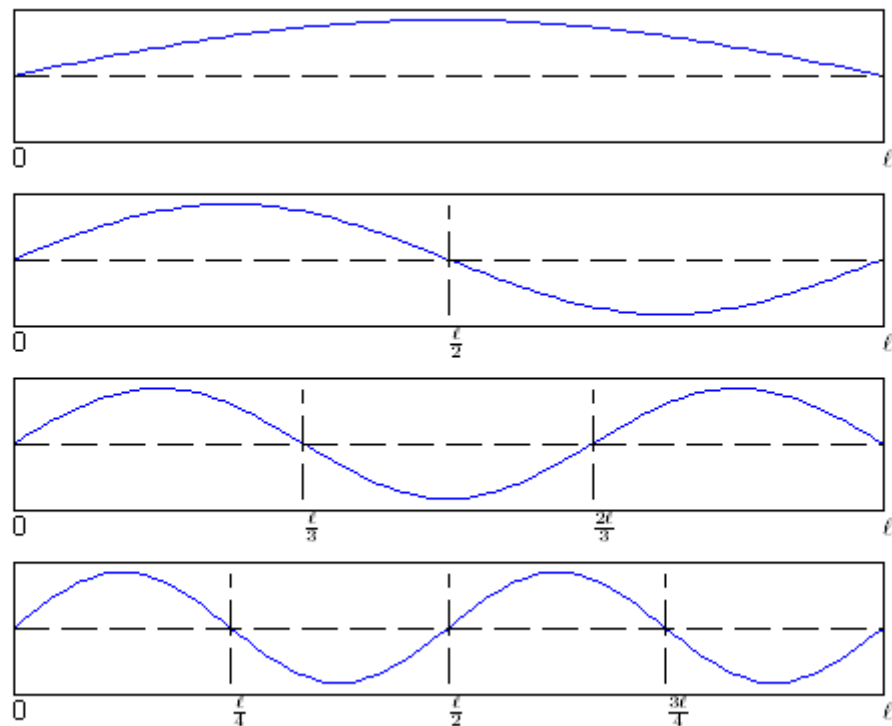


Figure 3.2 Shape of the first four modes of vibration for the axial vibrations of a clamped-clamped beam with constant section

³ Actually, this function only defines the *shape* of the vibration component, not the amplitude that will depend upon the particular initial conditions imposed to the system